

Group: 5

Team Members:

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Assignment: Requirements Engineering #2

User classes

Name	Number of users	Descriptions
Users (Primary Users)	Any person who installs the app	Users are the primary users of the system. They join accountability groups with friends and participate in daily challenges such as fitness, studying, or habit-building. Users will use the system several times per day to submit proof of completed tasks, review group progress, and track their streaks. Users may also review other members' submissions and verify whether challenge requirements were completed.
Group owner / Admins	Varies by group	A group owner is a user who creates a new accountability group. The group owner can invite members, define challenge rules, and manage deadlines for proof submissions. This role is not permanent, as any user in the system can create and manage their own group.

Elicitation Techniques

We used Synthetic Personas, combined with a facilitator agent, as our primary Elicitation technique. The following questionnaire was created by us for our Synthetic Persona's:

1. Could you each share a bit about your experiences with habit tracking or accountability apps? What's worked for you and what's been your biggest frustration?
2. When you think about proving you've completed a goal or habit, what would feel both satisfying and not too burdensome in your daily life?

3. What would motivate you to open an accountability app every day for several months? What features or experiences would keep you engaged?
4. How do you currently involve others in your goals? Do you prefer accountability to be private, social, or somewhere in between?

The facilitator agent then created some deep dive questions for each of our personas:

1. How might we balance different needs in one app? For example, what if there were optional competitive features that users could enable or disable? What specific settings would make this work for each of you?
2. Sarah, you mentioned leaderboards as motivating. Could you elaborate on what makes a good leaderboard? And Jessica and Marcus, is there any competitive element that might feel supportive rather than stressful to you?

Proof of Completion Tensions

You've all mentioned different preferences for proving goal completion.

3. What verification method would feel both meaningful and manageable in your daily routine? Sarah prefers video proof, Marcus likes screen recordings, and Jessica wants something simple.
4. How important is the ability to customize verification methods based on the type of goal? For instance, would you want different proof options for fitness goals versus meditation or studying?

Flexibility vs. Consistency

I'm noticing Marcus values streaks highly while Jessica needs flexibility for her unpredictable schedule.

5. What would an ideal "grace period" or flexibility feature look like that wouldn't undermine the streak motivation that Marcus values?
6. Marcus, is there a point where maintaining a streak becomes more stressful than motivating? And Jessica, what minimum level of consistency would still feel meaningful to you?

Notification and Reminder Preferences

7. When and how would you want to receive notifications from the app? Would you prefer fixed times, dynamic reminders based on your schedule, or something else?

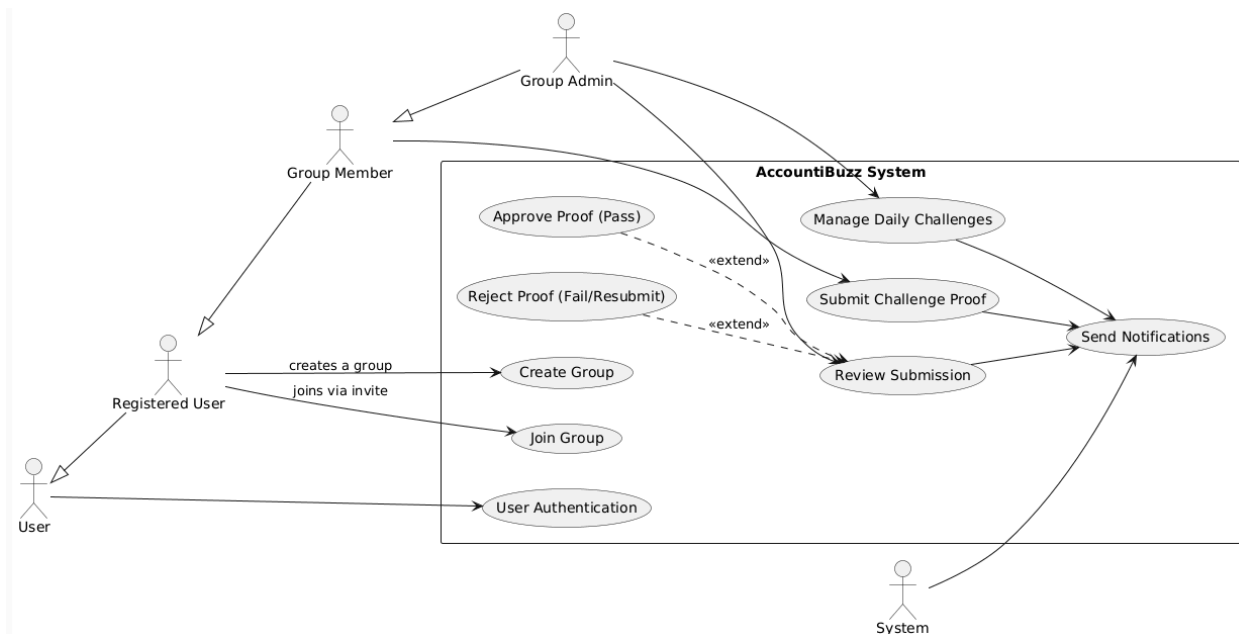
8. How would you feel about accountability partners being able to send you direct reminders? What boundaries would you want around this feature?

Group Dynamics

9. How would you want to organize different accountability groups? Sarah mentioned wanting multiple fitness-focused groups, Marcus has his dissertation group, and Jessica has her Self-Care Sisters.

10. What privacy controls would be important to you when sharing your progress with different groups?

Use Case Diagrams:



Use Case Specification

UC-000 User Authentication

ID and Name	UC-000 User Authentication		
Created By	Timothy Do, Chris Kelamyan	Date Created	March 6, 2026
Primary Actor	User	Secondary Actors	System
Description	This use case describes how a user creates an account and logs into the AccountiBuzz system. The user registers using their email and password, completes basic profile setup, and gains access to the application features such as joining groups and submitting challenge proof.		
Trigger	A user opens the application and selects Register or Sign In.		
Preconditions	PRE-1. The system is available and connected to the database. PRE-2. The user has access to the application interface.		
Postconditions	POST-1. The user account is created or authenticated successfully. POST-2. The user gains access to the application dashboard. POST-3. The user becomes a Registered User within the system.		
Normal Flow	1. The user opens the application. 2. The system displays login and registration options. 3. The user selects Register or Sign In. 4. The user enters required credentials. 5. The system validates the provided information. 6. The system creates a new account or authenticates the user. 7. The system redirects the user to the main dashboard.		
Alternative Flows	A1. User chooses to log in instead of registering 1. The user selects Sign In. 2. The user enters existing credentials. 3. The system verifies the credentials and grants access.		

Exceptions	E1. Invalid credentials 1. The user enters incorrect login information. 2. The system displays an error message. 3. The user must attempt login again.
Priority	High
Frequency of Use	High
Business Rules	BR-1. Each user account must have a unique email address. BR-2. Only authenticated users can access system features.
Special Requirements	SR-1. Login and registration should complete within a few seconds. SR-2. The system must securely store user credentials.
Assumptions	Users must create an account before joining groups or submitting challenges.

UC-001 Create Group

ID and Name	UC-001 Create Group		
Created By	Timothy Do, Chris Kelamyan	Date Created	March 6, 2026
Primary Actor	Registered User	Secondary Actors	System
Description	Describes how a registered user creates a new accountability group in AccountiBuzz. The user enters the group information, selects the group mode, and confirms the request. After successful creation, the system automatically assigns the creator as the group owner and admin.		
Trigger	A registered user selects the "Create Group" option.		
Preconditions	PRE-1. The user is registered and logged into the system. PRE-2. The user has completed basic profile setup. PRE-3. The system is available and connected to the database.		
Postconditions	POST-1. A new group is created and saved in the system. POST-2. The creator is assigned as the group owner and admin.		

	POST-3. The creator is redirected to the group's home page.
Normal Flow	8. The user navigates to the group creation screen. 9. The system displays the group creation form. 10. The user enters the group name. 11. The user enters the challenge description. 12. The user selects the daily submission deadline. 13. The user selects the group mode: Competitive or Collaborative. 14. The user confirms the group creation request. 15. The system validates the entered information. 16. The system creates the group record. 17. The system assigns the creator as the group owner and admin. 18. The system saves the group settings. 19. The system redirects the user to the group dashboard.
Alternative Flows	A1. User cancels group creation 1. The user selects 'Cancel' before confirming the request. 2. The system discards the entered information. 3. The system returns the user to the previous screen. A2. User changes selected mode before submission 1. The user reviews the available group modes. 2. The user changes the selected mode. 3. The system updates the form with the new selection.
Exceptions	E1. Missing information 1. The user submits the form with missing required fields. 2. The system displays an error message identifying the missing information. 3. The group is not created. E2. System or database error 1. The user submits the completed form. 2. The system attempts to create the group but fails. 3. The system displays an error message. 4. The group is not created.
Priority	High
Frequency of Use	Medium to High
Business Rules	BR-1. Every new group must have exactly one initial owner. BR-2. The creator is automatically assigned as the initial admin.

	BR-3. Each group must have a defined mode: Competitive or Collaborative.
Special Requirements	SR-1. The system should clearly explain the difference between Competitive and Collaborative mode before confirmation. SR-2. Group creation should complete within a few seconds under normal conditions.
Assumptions	• A registered user is allowed to create more than one group. • Invite links and member invitations are handled in a separate use case.

UC-002 Join Group

ID and Name	UC-002 Join Group		
Created By	Timothy Do, Chris Kelamyan	Date Created	March 6, 2026
Primary Actor	Registered User	Secondary Actors	System
Description	Describes how a registered user joins an existing accountability group using a valid invite link. After joining successfully, the user becomes a 'Group Member' and gains access to the group dashboard and active challenges.		
Trigger	A registered user opens a valid group invite link.		
Preconditions	PRE-1. The user is registered and logged into the system. PRE-2. The invite link is valid and active. PRE-3. The target group exists in the system. PRE-4. The system is available and connected to the database.		
Postconditions	POST-1. The user is added to the group's member list. POST-2. The user becomes a Group Member for that group. POST-3. The user gains access to the group dashboard and challenges information.		
Normal Flow	1. The user opens the invite link. 2. The system verifies that the invite link is valid. 3. The system displays the group information page.		

	- The user selects the 'Join Group' option. - The system checks whether the user is already a member of the group. - The system adds the user to the group's member list. - The system confirms successful group membership. - The system redirects the user to the group dashboard.
Alternative Flows	A1. User already belongs to the group - The user opens the invite link. - The system detects that the user is already a member. - The system informs the user that they already belong to the group. - The system redirects the user to the group dashboard.
Exceptions	E1. Invalid or expired invite link - The user opens the invite link. - The system determines that the invite link is invalid or expired. - The system displays an error message. - The user cannot join the group.
Priority	High
Frequency of Use	Medium Group joining typically occurs when new members are invited to participate.
Business Rules	BR-1. Only registered users can join a group. BR-2. A valid invite link is required to join a group. BR-3. A user cannot join the same group more than once. BR-4. A user may belong to multiple groups.
Special Requirements	SR-1. The system must validate invite links before allowing a user to join a group. SR-2. The system must prevent duplicate membership in the same group.
Assumptions	- Invite links are generated by the group owner or admin. - Invite links may expire or be disabled by the group owner.

UC-003 Submit Challenge Proof

ID and Name	UC-003 Submit Challenge Proof		
Created By	Timothy Do, Chris Kelamyan	Date Created	March 6, 2026
Primary Actor	Group Member	Secondary Actors	System
Description	Describes how a Group Member submits proof of completing a daily challenge in AccountiBuzz. The member uploads proof, such as a photo or video, before the submission deadline. After submission, the system records the proof and makes it available for admin review.		
Trigger	A Group Member selects the ‘Submit Challenge Proof’ option for the current daily challenge.		
Preconditions	PRE-1. The user is registered and logged into the system. PRE-2. The user is a Group Member of the target group. PRE-3. A challenge submission period is currently active. PRE-4. The system is available and connected to the database.		
Postconditions	POST-1. The proof submission is stored in the system. POST-2. The submission is associated with the member and the current challenge. POST-3. The submission is made available for review by the Group Admin.		
Normal Flow	1. The member opens the group dashboard. 2. The member navigates to the current daily challenge. 3. The member selects ‘Submit Challenge Proof’. 4. The system displays the proof submission interface. 5. The member uploads proof of completion, such as a photo or video. 6. The member confirms the submission. 7. The system validates the uploaded file. 8. The system records the proof submission. 9. The system marks the submission as pending review. 10. The system confirms successful submission to the member.		
Alternative Flows	A1. Member edits submission before confirmation		

	1. The member uploads proof. 2. The member reviews the uploaded content. 3. The member replaces the file before confirming submission. 4. The system updates the uploaded proof before final submission.
Exceptions	E1. Submission deadline has passed 1. The member attempts to submit proof after the deadline. 2. The system detects that the submission period has ended. 3. The system displays a message indicating that submissions are closed. 4. The submission is not accepted. E2. File upload fails 1. The member uploads proof. 2. The system detects a file upload failure. 3. The system displays an error message. 4. The member must attempt the upload again.
Priority	High
Frequency of Use	High Proof submissions occur daily as users complete their challenges.
Business Rules	BR-1. Only Group Members can submit challenge proof. BR-2. Each member may submit only one proof per challenge period unless resubmission is requested. BR-3. Proof submissions must occur before the daily deadline. BR-4. Submitted proof must be associated with a valid group challenge.
Special Requirements	SR-1. The system must support photo and video uploads. SR-2. Proof submissions should complete within a few seconds under normal network conditions.
Assumptions	• Members are responsible for ensuring that uploaded proof accurately represents completion of the challenge. • Submitted proof will be reviewed by the Group Admin.

UC-004 Review Submission

ID and Name	UC-004 Review Submission		
Created By	Timothy Do, Chris Kelamyan	Date Created	March 6, 2026
Primary Actor	Group Admin	Secondary Actors	System
Description	This use case describes how a Group Admin reviews a submitted proof in AccountiBuzz. After reviewing the uploaded proof, the admin may approve the proof as a pass or reject the proof and require resubmission. The system records the review result and updates the submission status.		
Trigger	A Group Admin opens a proof submission that is pending review.		
Preconditions	PRE-1. The user is registered and logged into the system. PRE-2. The user is the Group Admin for the target group. PRE-3. A proof submission exists and is pending review. PRE-4. The system is available and connected to the database.		
Postconditions	POST-1. The review decision is recorded in the system. POST-2. The submission status is updated. POST-3. The member is notified of the review result.		
Normal Flow	1. The admin opens the group dashboard. 2. The admin navigates to the list of pending submissions. 3. The system displays submissions awaiting review. 4. The admin selects a submission to review. 5. The system displays the submitted proof. 6. The admin reviews the proof. 7. The admin selects 'Approve Proof'. 8. The system records the approval decision. 9. The system marks the submission as approved. 10. The system updates the member's challenge status. 11. The system notifies the member of the approval result.		
Alternative Flows	A1. Reject proof and request resubmission 1. The admin reviews the submitted proof. 2. The admin selects 'Reject Proof'. 3. The system records the rejection decision. 4. The system marks the submission as rejected.		

	5. The system notifies the member that resubmission is required.
Exceptions	E1. Submission already reviewed 1. The admin opens a submission that has already been reviewed. 2. The system detects that the review is already complete. 3. The system displays the existing review result. 4. No additional review is allowed.
Priority	High
Frequency of Use	High Review occurs regularly as Group Admins review daily proof submissions.
Business Rules	BR-1. Only the Group Admin can review submissions for that group. BR-2. A submission must receive one review decision: approved or rejected. BR-3. A rejected submission may require resubmission if the deadline still allows it. BR-4. Review results must be recorded by the system.
Special Requirements	SR-1. The system must clearly display submitted proof before review. SR-2. Review actions should update the submission status immediately.
Assumptions	• The Group Admin reviews submissions honestly and according to group rules. • The system records review decisions for accountability and transparency.

UC-005 Manage Daily Challenges

ID and Name	UC-005 Manage Daily Challenges		
Created By	Timothy Do, Chris Kelamyan	Date Created	March 6, 2026
Primary Actor	Group Admin	Secondary Actors	System
Description	This use case describes how a Group Admin creates or manages daily challenges within an accountability group. The admin defines the		

	challenge description, submission rules, and submission deadline for group members.
Trigger	A Group Admin selects the ‘Manage Daily Challenge’ option from the group dashboard.
Preconditions	PRE-1. The user is registered and logged into the system. PRE-2. The user is the Group Admin for the group. PRE-3. The group exists in the system. PRE-4. The system is connected to the database.
Postconditions	POST-1. A challenge is created or updated in the system. POST-2. The challenge becomes visible to all group members. POST-3. The system records the challenge deadline and rules.
Normal Flow	1. The admin opens the group dashboard. 2. The admin selects Manage Daily Challenges. 3. The system displays the challenge configuration interface. 4. The admin enters the challenge description. 5. The admin sets the submission deadline. 6. The admin confirms the challenge creation. 7. The system validates the challenge information. 8. The system saves the challenge. 9. The challenge becomes visible to group members.
Alternative Flows	A1. Admin edits an existing challenge 1. The admin selects an existing challenge. 2. The admin updates the challenge description or deadline. 3. The system saves the updated challenge.
Exceptions	E1. Missing challenge information 1. The admin submits the challenge form with missing fields. 2. The system displays an error message. 3. The challenge is not created.
Priority	High
Frequency of Use	Medium Challenge configuration occurs occasionally when Group Admins create or update daily challenges for their group.

Business Rules	BR-1. Only a Group Admin can create or modify challenges. BR-2. Each group may have only one active daily challenge at a time. BR-3. All challenges must have a submission deadline.
Special Requirements	SR-1. The system should clearly display the active challenge to all group members. SR-2. Challenge updates should appear immediately to group members.
Assumptions	• The Group Admin is responsible for defining challenges that are clear and appropriate for the group. • Group members will follow the challenge rules defined by the admin. • The system allows only one active daily challenge per group at a time.

UC-006 Send Notifications

ID and Name	UC-006 Send Notifications		
Created By	Timothy Do, Chris Kelamyan	Date Created	March 6, 2026
Primary Actor	System	Secondary Actors	Group Member, Group Admin
Description	This use case describes how the system sends notifications to users regarding important events such as challenge deadlines, submission reminders, and review results. Notifications help users stay engaged and aware of group activity.		
Trigger	A system event occurs such as a challenge deadline approaching, a submission being reviewed, or a milestone being reached.		
Preconditions	PRE-1. The user is registered in the system. PRE-2. A relevant system event has occurred. PRE-3. The notification system is active.		
Postconditions	POST-1. A notification is delivered to the user. POST-2. The user is informed of relevant system activity.		
Normal Flow	1. A relevant system event occurs. 2. The system determines which users should receive notifications.		

	3. The system generates the notification message. 4. The system sends the notification to the appropriate users. 5. The user receives the notification.
Alternative Flows	A1. User disables notifications 1. The user disables notification preferences. 2. The system suppresses the notification.
Exceptions	E1. Notification delivery failure 1. The system attempts to send the notification. 2. The notification fails to deliver. 3. The system logs the failure and retries later.
Priority	Medium
Frequency of Use	High
Business Rules	BR-1. Notifications must be associated with a valid system event. BR-2. Users may control notification preferences.
Special Requirements	SR-1. Notifications should be delivered quickly after an event occurs.
Assumptions	Users rely on notifications to stay informed about challenge activity.

User Stories:

Link to JIRA

<https://comp586.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog>

All of our user stories for this project are maintained in Jira. The project backlog contains the complete set of stories referenced by these use cases.